Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8.	*(a)*	(i)		Enthalpy / energy change when 1 mol of substance is burned (1) Completely / in excess oxygen under standard conditions (1)	2			2		
		(ii)		C_2H_6 + **3½**O_2 → $2CO_2$ + $3H_2O$ (1) Bonds broken 1(C—C) + 3½(O=O) + 6(C—H) = 2080.5 + 6(C—H) (1) Bonds formed 4(C=O) + 6(O—H) = 5974 (1) 2080.5 + 6(C—H) – 5974 = –1561 (C—H) = 389 $kJmol^{-1}$ (1) award (3) for cao ecf possible		4		4	1 1 1	
		(iii)		Average used since each individual bond will be in a different environment and therefore have a different strength (1)			1	1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8.	*(b)*			**Indicative content** Correct in that energy produced per gram is 32.8 kJ from charcoal and 55.6 kJ from methane Both give CO_2 on burning 1 mol of each fuel produces 1 mol of CO_2 Wood for charcoal comes from (living) trees Methane comes from sources living millions of years ago / is a fossil fuel Charcoal is renewable / methane is non-renewable Trees take in CO_2 in photosynthesis Trees release the **same** amount of CO_2 on combustion that they took in during growth Charcoal burning is overall carbon neutral		1	5	6	1	
				***5*-6 marks** Must calculate energy per gram for both fuels. *The candidate constructs a relevant, coherent and logically structured account including key elements of the indicative content. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary are used accurately throughout.* **3- 4 marks** Clear comparison of methane and charcoal. *The candidate constructs a coherent account including many of the key elements of the indicative content. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary is generally sound.* **1-2 marks** Main focus on only one of methane or charcoal. *The candidate attempts to link at least two relevant points from the indicative content. Coherence is limited by omission and/or inclusion of irrelevant material. There is some evidence of appropriate use of scientific conventions and vocabulary.* **0 marks** *The candidate does not make any attempt or give an answer worthy of credit.*						
				Question 8 total	**2**	**5**	**6**	**13**	**4**	**0**